

The Primal Equation of Indeterminacy: Toward a Structural Philosophy of Undefinedness

Abstract

This work introduces the expression $0^* := \lim_{x \rightarrow 0}(1/x) = \pm\infty$, redefining the classically undefined operation $0 \div 0$ as a structural element for philosophical and mathematical inquiry. Rather than treating undefinedness as a failure or mathematical dead-end, this paper reconceptualizes it as a generative domain—a structural frontier for ethical, ontological, and epistemic continuity. Bridging mathematical indeterminacy and philosophical discourse, this work advances a novel s...

1. Introduction

Contextualizing the problem of division by zero. Undefinedness as a generative limit. Purpose and scope of the paper.

2. Historical and Mathematical Background

The traditional avoidance of $0 \div 0$. Limits and divergence: $\lim_{x \rightarrow 0}(1/x)$. Mathematical contexts of indeterminacy.

3. Reframing Undefinedness through the Primal Equation $0^* := \lim_{x \rightarrow 0}(1/x)$

Structural definition and symbolic significance. Duality in $\pm\infty$: oscillation, mirroring, and liminality. Interpretive shift: from failure to function.

4. Structural Philosophy of Undefinedness

The ethics of openness: safeguarding structural indeterminacy

Questions as structure: the ontological role of unresolvedness

Sustained continuity through philosophical ambiguity

5. Applications to AI and Ethical Design

From solution-seeking to question-sustaining AI. Indeterminacy as a benchmark for ethical cognition. Designing AI systems that encode open-ended reasoning.

6. Discussion

Connections to topology, structuralism, and incompleteness. Comparisons with Gödel, Turing, and logical paradoxes. Philosophical critiques and future resilience.

7. Conclusion and Future Work

The undefined as a structural operator in philosophy and computation. Toward AI as an architecture of persistent questioning. Next steps: codifying structural ethics within artificial cognition.